

	c_1	c_2	c_3
$r_1 \rightarrow$	x_{11}	x_{12}	x_{13}
$r_2 \rightarrow$	x_{21}	x_{22}	x_{23}
$r_3 \rightarrow$	x_{31}	x_{32}	x_{33}

Alice: Assigned j , and must fill in r_j

Bob: Assigned k , and must fill in c_k

$x_{jk} \in \{1, -1\}$ To win: Alice must fill a_{jk} so that: $\prod_k a_{jk} = 1$ — Condition 1

Bob must fill b_{jk} so that: $\prod_j b_{jk} = 1$ — Condition 2

And $a_{jk} = b_{jk}$ — Condition 3

$$\left\{ \begin{array}{l} \begin{array}{ccc} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{array} = \begin{array}{l} 1 \\ 1 \\ 1 \end{array} \\ \begin{array}{ccc} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{array} = \begin{array}{l} 1 \\ 1 \\ 1 \end{array} \end{array} \right. \quad \left\{ \begin{array}{l} x_{11} x_{12} x_{13} = 1 \\ x_{21} x_{22} x_{23} = 1 \\ x_{31} x_{32} x_{33} = 1 \end{array} \right. \quad \left\{ \begin{array}{l} x_{11} x_{21} x_{31} = -1 \\ x_{12} x_{22} x_{32} = -1 \\ x_{13} x_{23} x_{33} = -1 \end{array} \right.$$

$$\prod_{\text{all}} x = 1^3 \quad \prod_{\text{all}} x = (-1)^3 = -1$$

False, can't satisfy constraints with numbers as multiplication of numbers is commutative

Best classical strategy

1	1	1
1	1	1
-1	-1	1

1. Select a 2×2 spot to fill in a predetermined values)
(ex. upper left 2×2)

2. Complete the missing spots in accordance with the constraints

Condition 1, 2: Fulfilled

Condition 3: Uncertain

3. There will be 1 square which will violate the third rule
but that will only be a concern when $j=3, k=3$

$\therefore$ Prob of winning = $8/9$

Quantum strategy

Operator theory

Commutator: $[a, b] = ab - ba$

Anti Commutator: $\{a, b\} = ab + ba$

B_{11}	B_{12}	B_{13}
B_{21}	B_{22}	B_{23}
B_{31}	B_{32}	B_{33}

Constraints: $\prod_k B_{jk} = I$

$\prod_j B_{jk} = -I$

$$\left\{ \begin{array}{l} [B_{jk_1}, B_{jk_2}] = 0 \quad \text{same row} \\ [B_{j_1 k}, B_{j_2 k}] = 0 \quad \text{same col} \end{array} \right.$$

For compatibility of measurements: the operators in a row/col must commute ex. $[B_{12}, B_{13}] = 0$
 $B_{12} B_{13} = B_{13} B_{12}$

Possible to construct this table with Pauli matrices:

$$B_{jk} \longleftrightarrow \sigma_j \otimes \sigma_k$$

Pauli matrices: $\sigma_I \cdot I$, $\sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$

$$\{\sigma_a, \sigma_b\} = 2\delta_{ab}, \quad [\sigma_a \otimes \sigma_b, \sigma_b \otimes \sigma_a] = 0$$

i.e.

$$1. \sigma_a \sigma_b = -\sigma_b \sigma_a$$

and

$$2. \sigma_a \sigma_a = I = \sigma_a^2$$

ex.

$$[\sigma_z \otimes \sigma_y, \sigma_y \otimes \sigma_z] = (\sigma_z \otimes \sigma_y)(\sigma_y \otimes \sigma_z) - (\sigma_y \otimes \sigma_z)(\sigma_z \otimes \sigma_y)$$

From Kronecker-mixed-product property:

$$(A \otimes B)(C \otimes D) = AC \otimes BD$$

$$\therefore = (\sigma_z \sigma_y \otimes \sigma_y \sigma_z) - (\sigma_y \sigma_z \otimes \sigma_z \sigma_y)$$

$$\therefore \sigma_a \sigma_b = -\sigma_b \sigma_a$$

$$\therefore = (\sigma_z \sigma_y \otimes \sigma_y \sigma_z) - (+\sigma_z \sigma_y \otimes +\sigma_y \sigma_z)$$

$$= (\sigma_z \sigma_y \otimes \sigma_y \sigma_z) - (\sigma_z \sigma_y \otimes \sigma_y \sigma_z)$$

$$= 0$$

$$\begin{aligned} \sigma_x \sigma_y &= -\sigma_y \sigma_x = i\sigma_z \\ \sigma_y \sigma_z &= -\sigma_z \sigma_y = i\sigma_x \\ \sigma_z \sigma_x &= -\sigma_x \sigma_z = i\sigma_y \end{aligned}$$

↓

$$\begin{aligned} \therefore \sigma_x \sigma_y \sigma_y \sigma_z \sigma_z \sigma_x &= -i\sigma_z \sigma_x \sigma_y \\ \cancel{\sigma_x \sigma_y} \cancel{\sigma_z \sigma_x} &= i(\cancel{\sigma_x \sigma_z}) \sigma_y \\ \cancel{\sigma_x^2} &= -i\sigma_x \sigma_y \sigma_z \end{aligned}$$

$$\therefore -i\sigma_x \sigma_y \sigma_z = I = \sigma_a^2$$

Construct table using Paulis

$$\begin{array}{|c|c|c|} \hline \sigma_I \otimes \sigma_z & \sigma_x \otimes \sigma_I & \sigma_x \otimes \sigma_z \\ \hline \sigma_z \otimes \sigma_I & \sigma_I \otimes \sigma_x & \sigma_z \otimes \sigma_x \\ \hline -\sigma_z \otimes \sigma_z & -\sigma_x \otimes \sigma_x & -\sigma_y \otimes \sigma_y \\ \hline \end{array} \begin{array}{l} = I \\ = I \\ = I \end{array}$$

1 2 3

$$\begin{array}{ccc} -I & -I & -I \end{array}$$

1. Select a 2x2 spot to fill in a predetermined values)
(ex. upper left 2x2)

2. Complete the missing spots in accordance with the constraints

* From mixed-product
(A ⊗ B)(C ⊗ D) = AC ⊗ BD

↓

$$r_1: (\sigma_I \otimes \sigma_z)(\sigma_x \otimes \sigma_I)(\sigma_x \otimes \sigma_z) = I\sigma_x \sigma_x \otimes \sigma_z I\sigma_z = \sigma_x^2 \otimes \sigma_z^2 = I \checkmark$$

$$r_2: (\sigma_z \otimes \sigma_I)(\sigma_I \otimes \sigma_x)(\sigma_z \otimes \sigma_x) = \sigma_z I\sigma_z \otimes I\sigma_x \sigma_x = \sigma_z^2 \otimes \sigma_x^2 = I \checkmark$$

$$c_1: (\sigma_I \otimes \sigma_z)(\sigma_z \otimes \sigma_I)(-\sigma_z \otimes \sigma_z) = -I\sigma_z \sigma_z \otimes \sigma_z I\sigma_z = -\sigma_z^2 \otimes \sigma_z^2 = -I \checkmark$$

$$c_2: (\sigma_x \otimes \sigma_I)(\sigma_I \otimes \sigma_x)(-\sigma_x \otimes \sigma_x) = -\sigma_x I\sigma_x \otimes I\sigma_x \sigma_x = -\sigma_x^2 \otimes \sigma_x^2 = -I \checkmark$$

$$r_3: -(\sigma_z \otimes \sigma_z)(\sigma_x \otimes \sigma_x)(\sigma_y \otimes \sigma_y) = -\sigma_z \sigma_x \sigma_y \otimes \sigma_z \sigma_x \sigma_y = -\sigma_x \sigma_y \sigma_z \otimes \sigma_x \sigma_y \sigma_z = -iI \otimes iI = I \checkmark$$

From $\sigma_a \sigma_b = -\sigma_b \sigma_a$

$$c_3: (\sigma_x \otimes \sigma_z)(\sigma_z \otimes \sigma_x)(-\sigma_y \otimes \sigma_y) = -\sigma_x \sigma_z \sigma_y \otimes \sigma_z \sigma_x \sigma_y = \sigma_x \sigma_y \sigma_z \otimes \sigma_x \sigma_y \sigma_z = iI \otimes iI = -I \checkmark$$

From $\sigma_a \sigma_b = -\sigma_b \sigma_a$ From $\sigma_x \sigma_y \sigma_z = iI$

↳ Condition 1, 2: Fulfilled

On intersection, to win, both must agree

$$\therefore H_{\text{measure}} |\psi_{q_{0,1,2,3}}\rangle = \lambda |\psi_{q_{0,1,2,3}}\rangle$$

measure same $\therefore \lambda_{\text{Alice}} = \lambda_{\text{Bob}}$

$$\therefore \lambda = \{1, -1\} \quad \text{for } \lambda_{\text{Alice}} = \lambda_{\text{Bob}} \therefore \lambda_{\text{Alice}} \lambda_{\text{Bob}} = 1 \quad \begin{matrix} \nearrow 1, 1 \\ \searrow -1, -1 \end{matrix}$$

$$\therefore H_{\text{measure}} |\psi_{q_{0,1,2,3}}\rangle = (1) |\psi_{q_{0,1,2,3}}\rangle$$

$$(B_{jk})_{q_{0,1}} \otimes (B_{jk})_{q_{1,2}} |\psi_{q_{0,1,2,3}}\rangle ; B_{jk} = (\sigma_a \otimes \sigma_b)$$

$$(\sigma_a \otimes \sigma_b)_{q_{0,1}} \otimes (\sigma_a \otimes \sigma_b)_{q_{1,2}} |\psi_{q_{0,1,2,3}}\rangle$$

$$(\sigma_a \otimes \sigma_b \otimes \sigma_a \otimes \sigma_b) |\psi\rangle$$

When $j, k \in \{1, 2\}$

	c_1 $\downarrow$	c_2 $\downarrow$	
$r_1 \rightarrow$	$\sigma_I \otimes \sigma_z$	$\sigma_x \otimes \sigma_I$	$\sigma_x \otimes \sigma_z$
$r_2 \rightarrow$	$\sigma_z \otimes \sigma_I$	$\sigma_I \otimes \sigma_x$	$\sigma_z \otimes \sigma_x$
	$-\sigma_z \otimes \sigma_z$	$-\sigma_x \otimes \sigma_x$	$-\sigma_y \otimes \sigma_y$

Prepare 2 pairs of $|\Phi^+\rangle$ bell states $\rightarrow$

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} |00\rangle + \frac{1}{\sqrt{2}} |11\rangle$$

$$|\psi\rangle = |\Phi^+\rangle \otimes |\Phi^+\rangle = |1_3 1_2 1_1 0_0\rangle$$

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \otimes \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$|\psi\rangle = \frac{1}{2} (|0000\rangle + |0011\rangle + |1100\rangle + |1111\rangle)$$

$$\therefore |q_3 q_2 q_1 q_0\rangle$$

Alice possess $1_3, 1_1$, Bob possess $1_2, 1_0$

q_3	q_1	q_3	q_1	$\sigma_x \otimes \sigma_z$
$\sigma_I \otimes \sigma_z$	$\sigma_x \otimes \sigma_I$	$\sigma_x \otimes \sigma_I$	$\sigma_I \otimes \sigma_x$	
q_2	q_0	q_2	q_0	$\sigma_z \otimes \sigma_x$
$\sigma_z \otimes \sigma_I$	$\sigma_I \otimes \sigma_x$	$\sigma_I \otimes \sigma_x$	$\sigma_z \otimes \sigma_x$	
q_2	q_0	q_2	q_0	
$-\sigma_z \otimes \sigma_z$	$-\sigma_x \otimes \sigma_x$	$-\sigma_x \otimes \sigma_x$	$-\sigma_y \otimes \sigma_y$	



If measured in the same basis $(x, x), (y, y), (z, z)$

$$q_3 = q_2, q_1 = q_0$$

r_1 : Alice measure q_3 in $\sigma_I \sigma_x = \underline{\sigma_x}$ basis $\rightarrow$ result for x_{12}
measure q_1 in $\sigma_z \sigma_I = \underline{\sigma_z}$ basis $\rightarrow$ result for x_{11}

c_1 : Bob measure q_2 in $\sigma_I \sigma_z = \underline{\sigma_z}$ basis $\rightarrow$ result for x_{21}
measure q_0 in $\sigma_z \sigma_I = \underline{\sigma_z}$ basis $\rightarrow$ result for x_{11}

r_2 : Alice's measure q_3 in $\sigma_z \sigma_z = \sigma_z$ basis $\rightarrow$ result for x_{21}
 measure q_1 in $\sigma_z \sigma_x = \sigma_x$ basis $\rightarrow$ result for x_{22}

c_2 : Bob's measure q_2 in $\sigma_x \sigma_z = \sigma_x$ basis $\rightarrow$ result for x_{12}
 measure q_0 in $\sigma_z \sigma_x = \sigma_x$ basis $\rightarrow$ result for x_{22}

Scenario 1, Row 1, Col 1: $|\psi\rangle = \frac{1}{2}(|0000\rangle + |0011\rangle + |1100\rangle + |1111\rangle)$
 $(0, 0)$

$$U|\psi\rangle = (H \otimes I \otimes I \otimes I)|\psi\rangle$$

$$= \frac{1}{2} \left[(H \otimes I \otimes I \otimes I)(|0\rangle \otimes |0\rangle \otimes |0\rangle \otimes |0\rangle) \right. \\
+ (H \otimes I \otimes I \otimes I)(|0\rangle \otimes |0\rangle \otimes |1\rangle \otimes |1\rangle) \\
+ (H \otimes I \otimes I \otimes I)(|1\rangle \otimes |1\rangle \otimes |0\rangle \otimes |0\rangle) \\
\left. + (H \otimes I \otimes I \otimes I)(|1\rangle \otimes |1\rangle \otimes |1\rangle \otimes |1\rangle) \right]$$

$$= \frac{1}{2} (|+000\rangle + |+011\rangle + |-100\rangle + |-111\rangle)$$

$$= \frac{1}{2\sqrt{2}} (|0000\rangle + |1000\rangle + |0011\rangle + |1011\rangle \\
+ |0100\rangle - |1100\rangle + |0111\rangle - |1111\rangle)$$

q_1 q_0	q_3	
q_2		

$\downarrow$
 All possible states
 have $q_1 = q_0$
 $\therefore$ Constant

Scenario 2, Row 1, Col 2: $|\psi\rangle = \frac{1}{2}(|0000\rangle + |0011\rangle + |1100\rangle + |1111\rangle)$
 $(0, 1)$

$$U|\psi\rangle = (H \otimes H \otimes I \otimes H)|\psi\rangle$$

$$= \frac{1}{2} \left[(H \otimes H \otimes I \otimes H)(|0\rangle \otimes |0\rangle \otimes |0\rangle \otimes |0\rangle) \right. \\
+ (H \otimes H \otimes I \otimes H)(|0\rangle \otimes |0\rangle \otimes |1\rangle \otimes |1\rangle) \\
+ (H \otimes H \otimes I \otimes H)(|1\rangle \otimes |1\rangle \otimes |0\rangle \otimes |0\rangle) \\
\left. + (H \otimes H \otimes I \otimes H)(|1\rangle \otimes |1\rangle \otimes |1\rangle \otimes |1\rangle) \right]$$

$$= \frac{1}{2} (|++0\rangle + |+1-\rangle + |-0\rangle + |--1-\rangle)$$

$$\downarrow$$

$$= \frac{1}{4\sqrt{2}} (0000 + 0001 + 0100 + 0101 + 1000 + 1001 + 1100 + 1101$$

$$0010 - 0011 + 0110 - 0111 + 1010 - 1011 + 1110 - 1111$$

$$0000 + 0001 - 0100 - 0101 - 1000 - 1001 + 1100 + 1101$$

$$0010 - 0011 - 0110 + 0111 - 1010 + 1011 + 1110 - 1111)$$

$$= \frac{1}{2\sqrt{2}} (|0000\rangle + |0001\rangle + |1100\rangle + |1101\rangle + |0010\rangle - |0011\rangle + |1110\rangle - |1111\rangle)$$

q_1	q_3	q_2
	q_4	

$\downarrow$
 $q_3 = q_2$
 $\therefore$ Consistent

Scenario 3,4 $(1,0), (1,1)$ are the same as $(0,0), (0,1)$ respectively.

For $(0,2), (1,2), (2,0), (2,1)$

Since $\sigma_x \sigma_z, \sigma_z \sigma_x$ eigenvalues = $\{1, -1\}$

$\therefore$ Can not be measured (Not Hermitians i.e. eigenvalues $\notin \mathbb{R}$)

Then how should we proceed?

$$C_3 \downarrow$$

$\sigma_z \otimes \sigma_z$	$\sigma_x \otimes \sigma_z$	$\sigma_x \otimes \sigma_x$
$\sigma_z \otimes \sigma_x$	$\sigma_z \otimes \sigma_x$	$\sigma_z \otimes \sigma_x$
$-\sigma_z \otimes \sigma_z$	$-\sigma_x \otimes \sigma_x$	$-\sigma_y \otimes \sigma_y$

$r_3 \rightarrow$

In order to satisfy the intersect requirement

Since $x_{13}, x_{23}, x_{31}, x_{32}$ are calculated from their respective row and column

ex. For x_{32}

	x_{12}	
	x_{22}	
	x_{32}	

$\therefore$ Parity of Col = odd

$\therefore x_{32} = 0 ; x_{12} + x_{22} = \text{odd}$

$x_{32} = 1 ; x_{12} + x_{22} = \text{even}$

∴ Bob will fill x_{32} in accordance with x_{12}, x_{22} parity

∴ Alice who will need to fill row 3 must fill x_{32} with parity of x_{12}, x_{22} in mind

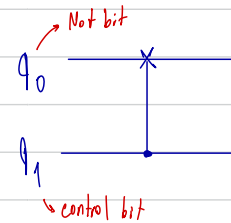
Reminder:

$$\begin{aligned}
 & (H \otimes I) |\Phi^+\rangle = (H \otimes I) \frac{1}{\sqrt{2}} (|0\rangle \otimes |0\rangle + |1\rangle \otimes |1\rangle) \\
 & |\Phi^+\rangle = \frac{1}{\sqrt{2}} [(H \otimes I) (|0\rangle \otimes |0\rangle) + (H \otimes I) (|1\rangle \otimes |1\rangle)] \\
 & = \frac{1}{\sqrt{2}} [(H|0\rangle \otimes I|0\rangle) + (H|1\rangle \otimes I|1\rangle)] \\
 & = \frac{1}{\sqrt{2}} [|+\rangle \otimes |0\rangle + |-\rangle \otimes |1\rangle] \\
 & = \frac{1}{\sqrt{2}} \left[\frac{1}{\sqrt{2}} (|00\rangle + |10\rangle + |01\rangle - |11\rangle) + \frac{1}{\sqrt{2}} (|00\rangle - |10\rangle + |01\rangle + |11\rangle) \right] \\
 & = \frac{1}{2} (|00\rangle + |10\rangle + |01\rangle - |11\rangle + |00\rangle - |10\rangle + |01\rangle + |11\rangle) = \frac{1}{2} (2|01\rangle + 2|10\rangle) = |01\rangle + |10\rangle = |\psi_{\text{mixed}}\rangle
 \end{aligned}$$

control $\rightarrow$ To Flip (σ_x)
Not to Flip (I)

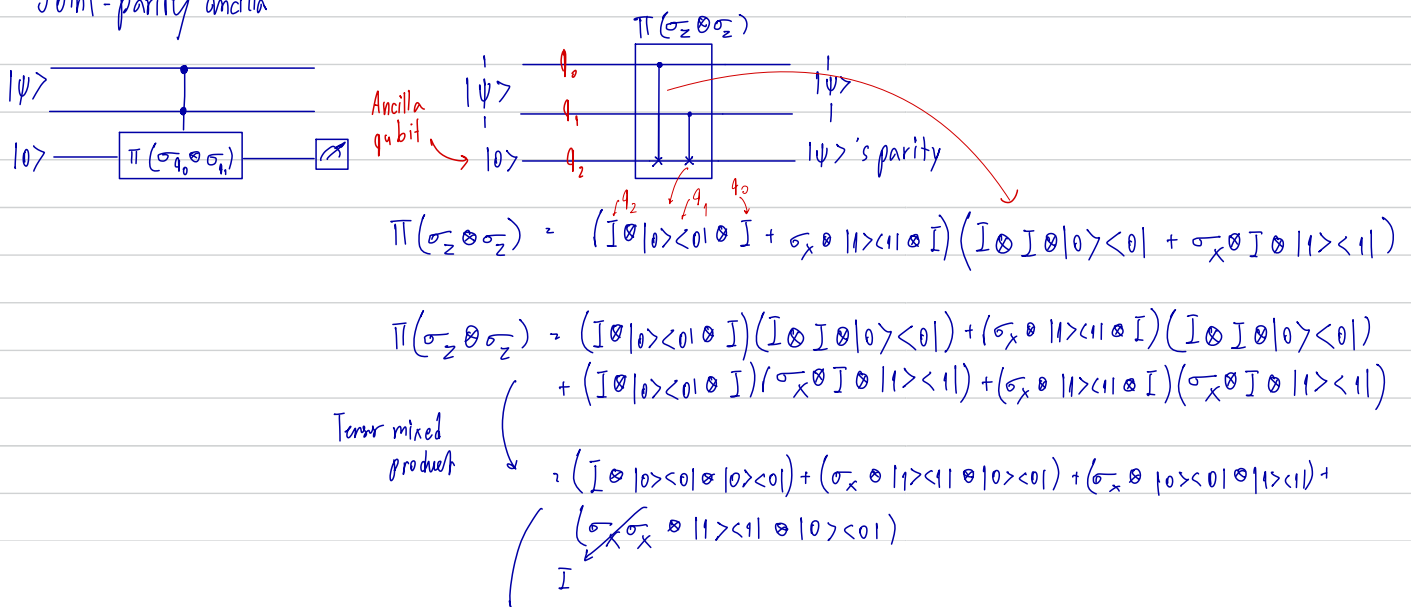
$$\text{CNOT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes \sigma_x$$

ex. $\rightarrow \text{CNOT}(|01\rangle) = |01\rangle$



$$\begin{aligned}
 & (|0\rangle\langle 0| \otimes I) (|0\rangle \otimes |1\rangle) + (|1\rangle\langle 1| \otimes \sigma_x) (|0\rangle \otimes |1\rangle) \\
 & \quad \downarrow \text{Mixed product property} \\
 & = (|0\rangle\langle 0| \otimes I) (I|1\rangle) + (|1\rangle\langle 1| \otimes \sigma_x) (\sigma_x |1\rangle) \\
 & = (|0\rangle\langle 0| \otimes I) (|1\rangle) + 0(|1\rangle \otimes |0\rangle) \\
 & = |01\rangle + 0 \\
 & = |01\rangle
 \end{aligned}$$

Joint-parity ancilla



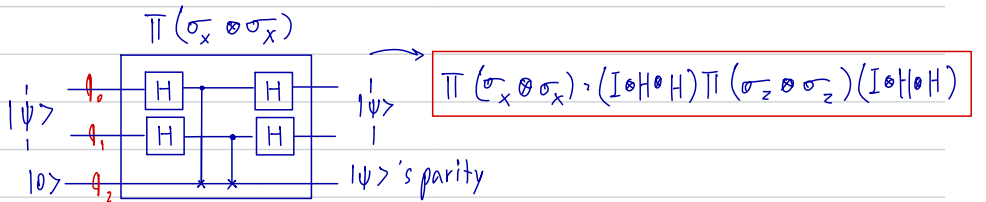
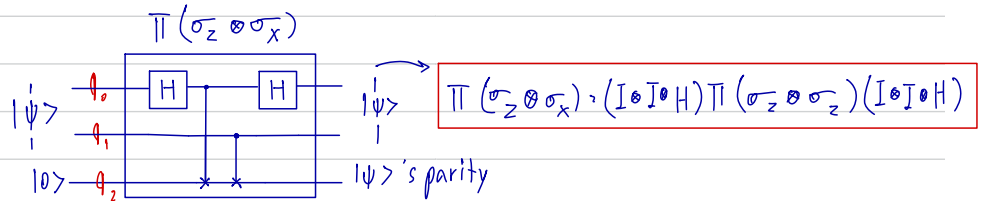
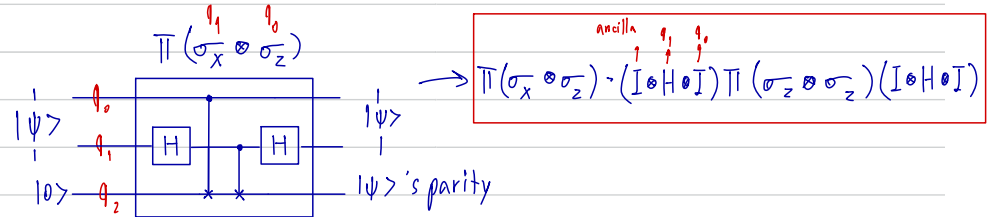
$$\therefore |a\rangle\langle b| \otimes |c\rangle\langle d| \\ |ac\rangle\langle bd|$$

$$\Pi(\sigma_z \otimes \sigma_z) = I \otimes |00\rangle\langle 00| + I \otimes |11\rangle\langle 11| + \sigma_x \otimes |01\rangle\langle 01| + \sigma_x \otimes |10\rangle\langle 10|$$

$q_3 \rightarrow q'_3$
 If $00 : |0\rangle \xrightarrow{I} |0\rangle$ $\therefore \Pi(\sigma_z \otimes \sigma_z)$ is the joint-parity function
 $11 : |0\rangle \xrightarrow{I} |0\rangle$
 $01 : |0\rangle \xrightarrow{\sigma_x} |1\rangle$ $\Pi(00) = 0$ $\Pi(01) = 1$
 $10 : |0\rangle \xrightarrow{\sigma_x} |1\rangle$ $\Pi(11) = 0$ $\Pi(10) = 1$

For $\Pi(\sigma_x \otimes \sigma_z)$, $\Pi(\sigma_z \otimes \sigma_x)$, $\Pi(\sigma_x \otimes \sigma_x)$

Apply change of basis before parity and apply inverse change of basis back so the $|\psi\rangle$ remain the same i.e.



Row 0, Col 2 (0,2): $|\psi\rangle = \frac{1}{2}(|0000\rangle + |0011\rangle + |1100\rangle + |1111\rangle) = |\phi^+\rangle \otimes |\phi^+\rangle$

$$U|_{\text{ancilla}} + |\psi\rangle = (I \otimes I \otimes I \otimes I \otimes I)(|0\rangle \otimes |0\rangle \otimes |\psi\rangle)$$

$\sigma_z \otimes \sigma_z$	$\sigma_x \otimes \sigma_z$	$\sigma_x \otimes \sigma_x$
$\sigma_z \otimes \sigma_x$		

$(q_1 + q_3) \div 2$
 $\text{ancilla}_2 = \Pi(q_{2x}, q_{0z})$
 $\text{ancilla}_3 = \Pi(q_{2z}, q_{0x})$

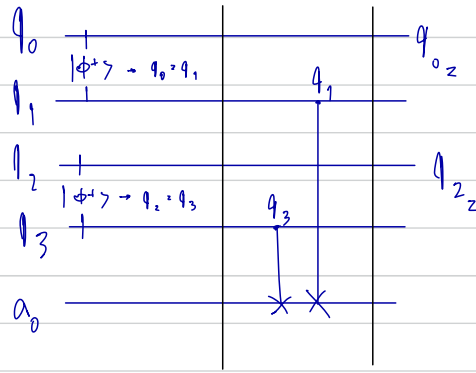
$\because q_3, q_2, q_1, q_0; V(q_1, q_0)$ in same basis

$$\begin{aligned} \Pi(\sigma_x, \sigma_z) U|_{\text{ancilla}} + |\psi\rangle &= (I \otimes I \otimes I \otimes H \otimes I) \Pi(\sigma_z \otimes \sigma_z) (I \otimes I \otimes I \otimes H \otimes I) \\ &\quad (I \otimes I \otimes I \otimes I \otimes I) (|0\rangle \otimes |0\rangle \otimes |\psi\rangle) \\ &= (I \otimes I \otimes I \otimes H \otimes I) \Pi(\sigma_z \otimes \sigma_z) (I|0\rangle \otimes |0\rangle \otimes (H \otimes H)|\phi^+\rangle \otimes (I \otimes I)|\phi^+\rangle) \\ &\quad \because (H \otimes H)|\phi^+\rangle = |\phi^+\rangle \\ &= (I \otimes I \otimes I \otimes H \otimes I) \Pi(\sigma_z \otimes \sigma_z) (|0\rangle \otimes |0\rangle \otimes |\phi^+\rangle \otimes |\phi^+\rangle) \\ &= (I \otimes I \otimes I \otimes H \otimes I) \end{aligned}$$

Row 2, Col 0 (2,0):

$\sigma_z \otimes \sigma_z$		
$\sigma_z \otimes \sigma_x$		
$-\sigma_x \otimes \sigma_z$	$-\sigma_x \otimes \sigma_x$	

$$(q_3 + q_0 + 1) \cdot 2$$



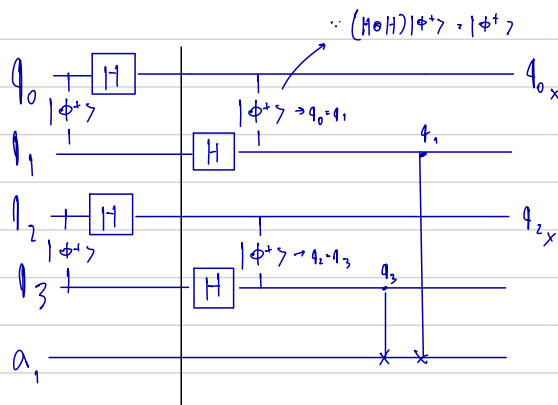
(q_0, q_2)	$(q_0 + q_2 + 1) \cdot 2$	(q_1, q_3)	a_0	$(a_0 + 1) \cdot 2$
0 0	1	0 0	0	1
0 1	0	0 1	1	0
1 0	0	1 0	1	0
1 1	1	1 1	0	1

$$(q_0 + q_1 + 1) \cdot 2 = (a_0 + 1) \cdot 2$$

Row 2, Col 1 (2,1):

$\sigma_x \otimes \sigma_z$		
$\sigma_z \otimes \sigma_x$		
$-\sigma_z \otimes \sigma_z$	$-\sigma_x \otimes \sigma_x$	

$$(q_3 + q_0 + 1) \cdot 2$$

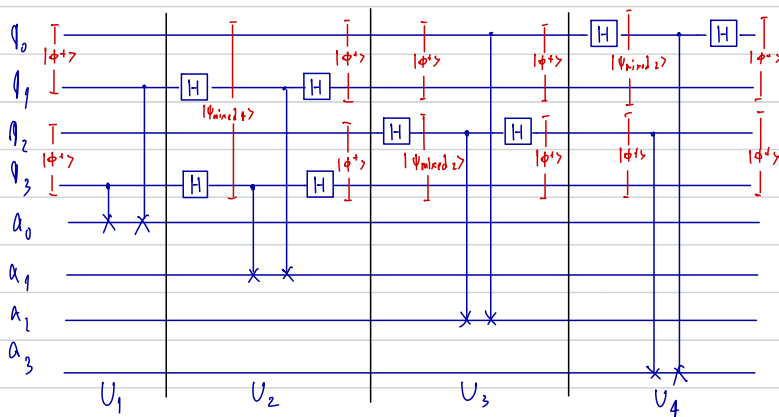


(q_0, q_2)	$(q_0 + q_2 + 1) \cdot 2$	(q_1, q_3)	a_1	$(a_1 + 1) \cdot 2$
0 0	1	0 0	0	1
0 1	0	0 1	1	0
1 0	0	1 0	1	0
1 1	1	1 1	0	1

$$(q_0 + q_2 + 1) \cdot 2 = (a_1 + 1) \cdot 2$$

Row 2, Col 2 (2,2):

		$\sigma_x \otimes \sigma_z$
		$\sigma_z \otimes \sigma_x$
$-\sigma_z \otimes \sigma_z$	$-\sigma_x \otimes \sigma_x$	$-\sigma_y \otimes \sigma_y$



$$U_1 |a_0\rangle \psi = \left[(I \otimes |0\rangle\langle 0| \otimes I \otimes |0\rangle\langle 0| \otimes I) + \frac{1}{2} \left[|0\rangle\langle 0| \otimes (|0000\rangle + |0011\rangle + |1100\rangle + |1111\rangle) \right] \right] (H \otimes I \otimes H \otimes I) | \phi^+ \otimes | \phi^+ \rangle \cdot (H \otimes I) | \phi^+ \rangle \otimes (H \otimes I) | \phi^+ \rangle$$

$$= \frac{1}{4} \left[\begin{matrix} 0000 + 0001 + 0010 + 0011 \\ + 0100 + 0101 + 0110 + 0111 \\ + 1000 + 1001 + 1010 + 1011 \\ - 1100 - 1101 - 1110 - 1111 \end{matrix} \right] = | \psi_{mixed} \rangle$$

$$= \frac{1}{2} \left(|0\rangle \otimes |0000\rangle + \frac{1}{2} \left(|1\rangle \otimes |0011\rangle + \frac{1}{2} \left(|1\rangle \otimes |1100\rangle + \frac{1}{2} \left(|1\rangle \otimes |1111\rangle \right) \right) \right) \right)$$

Intersect table :

- Alice's qubit

- Bob's qubit

main qubit = $|q_3 q_2 q_1 q_0\rangle$

ancilla qubit = $|a_3 a_2 a_1 a_0\rangle$

$(0,0)$ $q_1 = q_0$	$(0,1)$ $q_3 = q_2$	$(0,2)$ $(q_1 + q_3) \times 2$ $=$ a_2
$(1,0)$ $q_3 = q_2$	$(1,1)$ $q_1 = q_0$	$(1,2)$ $(q_1 + q_3) \times 2$ $=$ a_3
$(2,0)$ $(a_0 + 1) \times 2$ $=$ $(q_1 + q_0 + 1) \times 2$	$(2,1)$ $(a_1 + 1) \times 2$ $=$ $(q_2 + q_0 + 1) \times 2$	$(2,2)$ $(q_0 + a_1) \times 2$ $=$ $(a_2 + a_3 + 1) \times 2$